

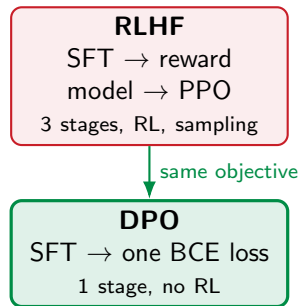
Alignment without the RL

Last time (GRPO) we made RL *cheaper* by deleting the critic. DPO asks a bolder question:

*What if we delete the reward model **and** the RL loop entirely?*

DPO shows the RLHF objective has a **closed-form** solution, and that inverting it turns preference alignment into a **single classification loss** – trained directly on preference pairs, no sampling, no PPO.

The title says it: *“Your language model is secretly a reward model.”* Today, why that is literally true.



Outline

Why alignment is hard

Preliminaries: the math of preferences

The DPO derivation

Understanding the update

Does it work?

Recap

Why is aligning an LLM hard?

The standard RLHF pipeline has a lot of moving parts.

The three-stage RLHF pipeline

RLHF: reward model + RL loop

2 models trained + RL loop + in-loop sampling



DPO: one classification loss

1 loss - no reward model - no RL - no sampling



RLHF (top): SFT $\rightarrow$ fit a reward model on human comparisons $\rightarrow$ optimize the policy with PPO against that reward, with a KL leash to the SFT model.

The pain: **multiple** large models (reward + policy + critic), **in-loop sampling**

Why preferences, and why a KL leash?

Why preferences over demonstrations?

- ▶ Humans are better at saying “*A is better than B*” than at writing the gold answer.
- ▶ Relative judgments are cheaper and more consistent to collect.

Why keep a KL constraint to π_{ref} ?

- ▶ Stay where the reward model is *accurate*.
- ▶ Preserve diversity; avoid collapsing onto one high-reward answer (reward hacking).

The RLHF objective:

$$\max_{\pi_{\theta}} \mathbb{E}_{x, y \sim \pi_{\theta}} [r_{\varphi}(x, y)] - \beta \mathbb{D}_{\text{KL}}[\pi_{\theta} \parallel \pi_{\text{ref}}]$$

Maximize reward, but do not stray far from the reference.

DPO will optimize *exactly* this objective – just without ever running RL.

The math of preferences

Bradley-Terry, and the reward objective we want to solve.

Bradley-Terry: turning preferences into probabilities

A preference dataset is triples (x, y_w, y_l) : prompt, **preferred**, **dispreferred**. The Bradley-Terry model says the chance y_1 beats y_2 is a softmax over rewards:

$$p^*(y_1 \succ y_2 \mid x) = \frac{\exp r(x, y_1)}{\exp r(x, y_1) + \exp r(x, y_2)} = \sigma(r(x, y_1) - r(x, y_2))$$

A reward model is fit by maximum likelihood – i.e. binary classification on which answer won:

$$\mathcal{L}_R = -\mathbb{E}_{(x, y_w, y_l)} \left[\log \sigma(r_\varphi(x, y_w) - r_\varphi(x, y_l)) \right]$$

Notice: only the **difference** of rewards ever appears. Hold onto this – it is the hinge of the whole derivation.

The derivation

From a KL-constrained objective to a classification loss, in four steps.

Step 1: the RLHF objective has a closed-form optimum

For *any* reward r , the policy that maximizes reward $-\beta\text{KL}$ is known in closed form:

$$\pi_r(y \mid x) = \frac{1}{Z(x)} \pi_{\text{ref}}(y \mid x) \exp\left(\frac{1}{\beta} r(x, y)\right) \quad Z(x) = \sum_y \pi_{\text{ref}}(y \mid x) \exp\left(\frac{1}{\beta} r(x, y)\right)$$

Why (sketch): rewrite the objective as $\min_{\pi} \mathbb{E}[\mathbb{D}_{\text{KL}}(\pi \parallel \pi^*)] - \log Z(x)$, where π^* is the boxed distribution. $Z(x)$ does not depend on π , so by Gibbs' inequality the KL is minimized (to 0) exactly when $\pi = \pi^*$. ■

But $Z(x) = \sum_y (\dots)$ sums over **all possible completions** – astronomically many token sequences. Eq. is exact but **intractable**. This is what forces classic RLHF into sampling-based RL.

Step 2: invert – the reward is a log-ratio

Take logs of the boxed optimum and solve for the reward:

$$r(x, y) = \beta \log \frac{\pi_r(y \mid x)}{\pi_{\text{ref}}(y \mid x)} + \beta \log Z(x)$$

- ▶ The reward that a policy is optimal for equals the β -**scaled log-ratio** of that policy to the reference ...
- ▶ ... plus a term $\beta \log Z(x)$ that depends on the *prompt only*, not on y .

This is the “secretly a reward model” identity: *any* language model π_θ **defines** an implicit reward $\hat{r}_\theta(x, y) = \beta \log \frac{\pi_\theta(y \mid x)}{\pi_{\text{ref}}(y \mid x)}$.

Step 3: the intractable $Z(x)$ cancels

Plug the reward identity into Bradley-Terry, which only sees reward *differences*:

$$r(x, y_w) - r(x, y_l) = \beta \log \frac{\pi(y_w | x)}{\pi_{\text{ref}}(y_w | x)} + \underbrace{\beta \log Z(x)}_{\text{prompt only}} - \beta \log \frac{\pi(y_l | x)}{\pi_{\text{ref}}(y_l | x)} - \underbrace{\beta \log Z(x)}_{\text{prompt only}}$$

The two $\beta \log Z(x)$ terms are **identical** and **cancel**. What was intractable becomes *irrelevant*.

Why it works: (a) both completions share the same prompt x , and (b) the preference model depends only on reward *differences*. Rewards differing by any prompt-only $f(x)$ give the same preferences – DPO just picks the representative with $Z(x) = 1$.

Step 4: the DPO loss

Substitute, and fit π_θ by maximum likelihood on the preference data:

$$\mathcal{L}_{\text{DPO}} = -\mathbb{E}_{(x, y_w, y_l)} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_\theta(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right) \right]$$

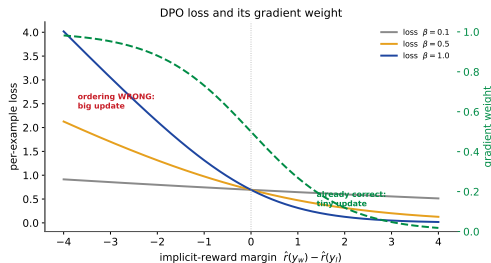
- ▶ y_w, y_l – preferred / dispreferred completion.
- ▶ π_θ – the policy we train; π_{ref} – frozen SFT model (never updated).
- ▶ β – KL strength / sigmoid temperature.

It is just **binary cross-entropy**: “make the preferred completion’s implicit reward exceed the dispreferred one’s.”

The gradient: focus on what you got wrong

$$\nabla_{\theta} \mathcal{L}_{\text{DPO}} = -\beta \mathbb{E} \left[\underbrace{\sigma(\hat{r}_{\theta}(x, y_I) - \hat{r}_{\theta}(x, y_W))}_{\text{weight: large when ranking is WRONG}} (\nabla_{\theta} \log \pi_{\theta}(y_W | x) - \nabla_{\theta} \log \pi_{\theta}(y_I | x)) \right]$$

- Push **up** y_W , push **down** y_I .
- Scale each pair by how badly the current *implicit* reward orders it.
- Pairs already ranked right barely move; mis-ranked pairs dominate the update.



Drop that weight (naive “raise y_W , lower y_I ”) and the model **degenerates** into garbage. The dynamic weight is the secret sauce.

The recipe in code

```
import torch.nn.functional as F

def dpo_loss(pi_logps, ref_logps, yw, yl,
             beta):
    # log-probs of the whole chosen /
    rejected sequences
    pi_w, pi_l = pi_logps[yw], pi_logps[yl]
    ref_w, ref_l = ref_logps[yw], ref_logps[yl]

    pi_logratio = pi_w - pi_l
    ref_logratio = ref_w - ref_l

    # binary cross-entropy on the implicit
    -reward margin
    loss = -F.logsigmoid(beta * (
        pi_logratio - ref_logratio))

    # implicit rewards (for logging only)
    rewards = beta * (pi_logps - ref_logps
        ).detach()
    return loss, rewards
```

What you need:

- ▶ preference triples (x, y_w, y_l)
- ▶ a frozen reference π_{ref} (= SFT model)

Per batch: four log-probs (policy & ref, on y_w & y_l) $\rightarrow$ sigmoid loss.

What you do not need: reward model, critic, policy sampling, RL tuning.

Defaults: $\beta = 0.1$, batch 64, lr 10^{-6} .

DPO vs RLHF, side by side

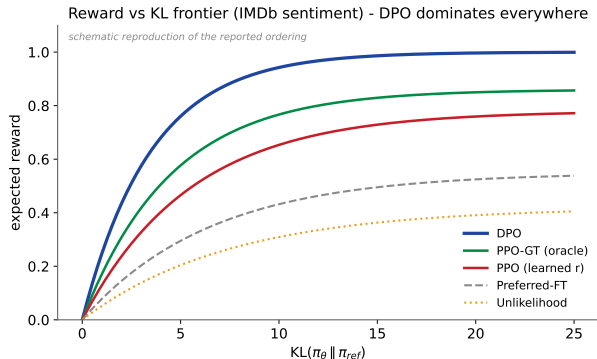
	RLHF (SFT→RM→PPO)	DPO
Explicit reward model	yes (separate net)	no – implicit $\hat{r} = \beta \log \frac{\pi_{\theta}}{\pi_{\text{ref}}}$
RL loop / critic	yes	no
Sample from policy in training	yes (on-policy)	no (offline)
Loss type	policy gradient (unstable)	classification / BCE (stable)
Objective optimized	KL-constrained reward max	the same , reparameterized

DPO is not an approximation – it optimizes *exactly* the RLHF objective, just written as a loss over the policy itself.

Does it work?

A computable frontier, win rates, and temperature robustness.

Reward vs KL: DPO dominates the frontier

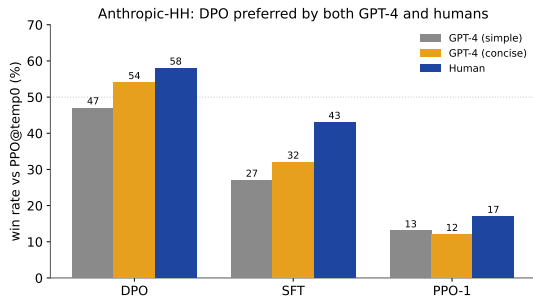
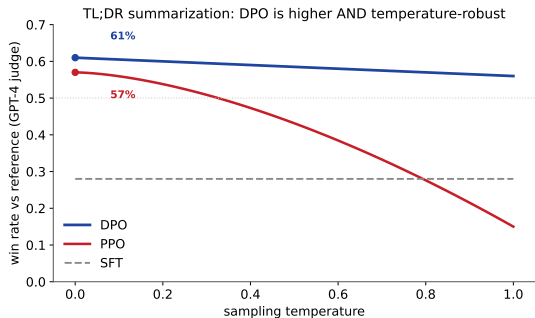


On IMDb the reward is a known sentiment classifier, so the exact reward-vs-KL trade-off is *computable*.

- DPO reaches **higher reward at every KL** budget.
- It even beats **PPO-GT** – PPO given the *ground-truth* reward.

Curves reproduce the reported ordering (schematic shape).

Win rates and temperature robustness



Summarization (left): DPO $\approx$ 61% win rate vs $\approx$ 57% for PPO – and DPO stays high as sampling temperature rises, while PPO collapses. **Dialogue (right):** on Anthropic-HH, DPO is preferred over PPO by *both* GPT-4 and human raters. GPT-4 agrees with humans about as often as humans agree with each other.

Recap and caveats

- ▶ DPO optimizes the **exact RLHF objective** with a single classification loss.
- ▶ Key identity: a language model *is* a reward model, $\hat{r}_\theta = \beta \log(\pi_\theta / \pi_{\text{ref}})$.
- ▶ The intractable partition function $Z(x)$ **cancels** because preferences use only reward differences within a prompt.
- ▶ No reward model, no critic, no policy sampling, minimal tuning – and it matches or beats PPO.
- ▶ The reference π_{ref} is **essential** (it defines the implicit reward); and the dynamic gradient weight is what prevents collapse.
- ▶ Caveat: DPO is **offline** – open questions remain on exploration and out-of-distribution behavior vs on-policy RL.

Next: *LoRA* – once we know *how* to fine-tune (SFT, DPO, GRPO), how do we do it **cheaply**, without a full copy of a 175B model per task?